

```
public class Student {  
  
    private String name;  
    private String id;  
    private String grade;  
  
    public Student(String name, String id, String grade) {  
        this.name = name;  
        this.id = id;  
        this.grade = grade;  
    }  
  
    public String getName() {  
        return name;  
    }  
  
    public void setName(String name) {  
        this.name = name;  
    }  
  
    public String getId() {  
        return id;  
    }  
  
    public void setId(String id) {  
        this.id = id;  
    }  
  
    public String getGrade() {  
        return grade;  
    }  
  
    public void setGrade(String grade) {  
        this.grade = grade;  
    }  
}
```

```
1 package designpatternsandprinciple.ques10;
2
3 public class StudentView {
4
5     public void displayStudentDetails(
6         String name,
7         String id,
8         String grade) {
9
10        System.out.println("Student Details");
11        System.out.println("Name   : " + name);
12        System.out.println("ID     : " + id);
13        System.out.println("Grade  : " + grade);
14    }
15 }
```

```
1 package designpatternsandprinciple.ques10;
2
3
4 public class StudentController {
5
6     private Student model;
7     private StudentView view;
8
9     public StudentController(
10         Student model,
11         StudentView view) {
12
13         this.model = model;
14         this.view = view;
15     }
16
17     public void setStudentName(String name) {
18         model.setName(name);
19     }
20
21     public String getStudentName() {
22         return model.getName();
23     }
24
25     public void setStudentGrade(String grade) {
26         model.setGrade(grade);
27     }
28
29     public void updateView() {
30
31         view.displayStudentDetails(
32             model.getName(),
33             model.getId(),
34             model.getGrade());
35     }
36 }
37
```



```
1 package designpatternsandprinciple.ques10;
2
3
4 public class MVCTest {
5
6     public static void main(String[] args) {
7
8         Student student =
9             new Student("Ambika", "101", "A");
10
11         StudentView view = new StudentView();
12
13         StudentController controller =
14             new StudentController(student, view);
15
16         System.out.println("Initial Student Details:");
17         controller.updateView();
18
19         System.out.println();
20
21         controller.setStudentName("Ambika Mishra");
22         controller.setStudentGrade("A+");
23
24         System.out.println("Updated Student Details:");
25         controller.updateView();
26     }
27 }
```

Initial Student Details:

Student Details

Name : Ambika

ID : 101

Grade : A

Updated Student Details:

Student Details

Name : Ambika Mishra

ID : 101

Grade : A+